

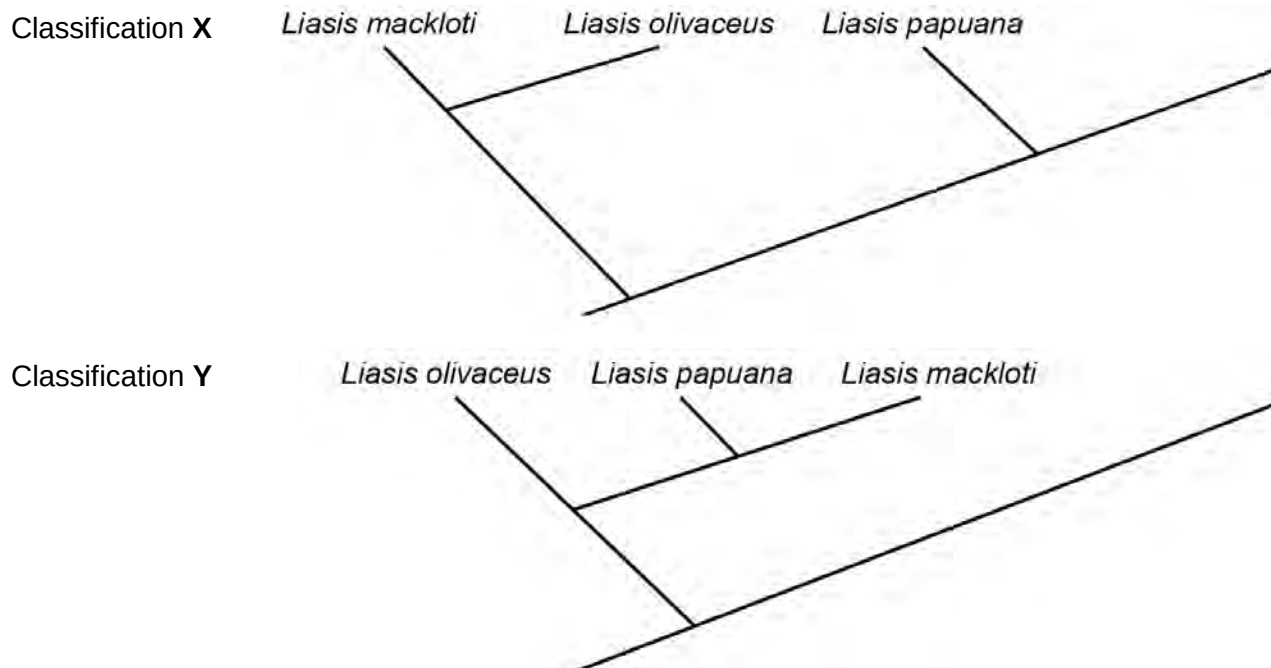
0 8

Figure 3 shows two different ways of classifying the same three species of snake.

- Classification **X** is based on the frequency of observable characteristics
- Classification **Y** is based on other comparisons of genetic characteristics.

All three species of snake belong to the Python family.

Figure 3



0 8 . 1

What do these classifications suggest about the evolutionary relationships between these species of snake?

[2 marks]

Classification **X** _____

Classification **Y** _____

0 8 . 2

Complete **Table 4** below to show the missing names of the taxa when classifying these snakes.

[1 mark]

Table 4

Taxon (hierarchical order)	Name
	Eukaryote
	Animal
	Chordata
	Reptilia
	Squamata
Family	Python

0 8 . 3

There is a debate about the name of one of these species of snake. Some scientists name it *Liasis papuana* and other scientists name it *Apodora papuana*.

Give the name of the taxon about which the scientists disagree.

[1 mark]

0 8 . 4

State **three** comparisons of genetic diversity that the scientists used in order to generate Classification Y.

[3 marks]

1

2

3

Turn over ►

